Functional vs. Non-functional Requirements

In the context of system design interviews, understanding functional and non-functional requirements is key to showcasing your ability to design a system that meets both the specific actions it should perform and how it should perform them.

Functional Requirements

- **Definition**: These are the requirements that define what a system is supposed to do. They describe the various functions that the system must perform.
- **Examples**:
  - A user authentication system must validate user credentials and provide access levels.
  - An e-commerce website should allow users to browse products, add them to a cart, and complete purchases.
  - A report generation system must collect data, process it, and generate timely reports.

Importance in Interviews

- **Demonstrates Understanding of Core Features**: Shows you know what the system needs to do to satisfy its primary objectives.
- **Basis for System Design**: Functional requirements often form the backbone of your system design.

Non-Functional Requirements

- **Definition**: These requirements describe how the system performs a task, rather than what tasks it performs. They are related to the quality attributes of the system.
- **Examples**:
  - Scalability: The system should handle growth in users or data.
  - Performance: The system should process transactions within a specified time.
  - Availability: The system should be up and running a defined percentage of time.
  - Security: The system must protect sensitive data and resist unauthorized access.

Importance in Interviews

- **Showcases Depth of Design Knowledge**: Demonstrates your understanding of the broader implications of system design.
- **Highlights System Robustness and Quality**: Reflects how well your system design can meet real-world constraints and user expectations.

Integrating Both in Interviews

- **Scenario-Based Discussions**: When presented with a scenario, identify both the functional (what the system should do) and non-functional (how the system should do it) requirements.
- **Balancing Act**: Exhibit your ability to balance both types of requirements, showing that you can design a system that not only meets its functional goals but also performs effectively, securely, and reliably.

In System Design Interviews

When you're in a system design interview, here's how you can handle these requirements:

1. **Clarify Requirements**: Start by asking questions to understand both functional and non-functional requirements. Interviewers often leave these vague to see if you'll probe for more details.
2. **Prioritize**: Not all requirements are equally important. Identify which ones are critical for the system's success.
3. **Trade-offs**: Discuss trade-offs related to different architectural decisions, especially concerning non-functional requirements. For example, a system highly optimized for read operations might have slower write operations.
4. **Use Real-World Examples**: If you can, relate your points to real-world systems or your past experiences. This shows practical understanding.
5. **Balance**: Ensure you're not focusing too much on one type of requirement over the other. A well-rounded approach is often necessary.

Remember, in system design interviews, interviewers are often interested in seeing how you think and approach problems, not just your final solution. Demonstrating a clear understanding of both functional and non-functional requirements is key to showing your comprehensive knowledge in system design.